

Analytical Geometry

Advanced Problem Set: Lines & Planes

Curated from JEE Advanced, AMC/HMMT, and IIT Competition Archives | + 2 Original Brilliant-Level Problems

This document contains 22 carefully selected problems in Analytical Geometry, divided into two main topics — Lines (2D) and Planes (3D). The first 10 questions in each topic were sourced from reputable competition archives and university entrance examinations (JEE Advanced, JEE Main, AMC 12, and HMMT). The final 2 questions are original compositions at Brilliant.org difficulty level.

- **Q1–Q10 per section:** Sourced from public competition archives (source listed in brackets).
 - ★ **Brilliant Q1–Q2 (end of document):** Original problems composed for this document, not found on the internet.
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SECTION 1 — LINES (2D Analytical Geometry)

All problems in this section involve straight lines in the Cartesian plane. They require reasoning about loci, envelopes, angle bisectors, concurrent families, and projections — far beyond routine equation-finding.

Problems from Top Institutions (Internet-Sourced)

Q1. Weighted distances along a transversal [JEE Advanced 2013]

A line through $A(-5, -4)$ meets the lines $L_1: x + 3y + 2 = 0$, $L_2: 2x + y + 4 = 0$, and $L_3: x - y - 5 = 0$ at points B , C , and D respectively.

If the following identity holds:

$$(15/AB)^2 + (10/AC)^2 = (6/AD)^2$$

find the equation of the line.

Q2. Locus from perpendicular-distance constraint [JEE Main — Coordinate Geometry PYQ]

A point P moves in the Cartesian plane such that the sum of its distances from two fixed perpendicular lines equals 1.

Determine the locus of P and identify the geometric figure it represents.
Describe all four regions of the plane and the equation in each.

Q3. Acute-angle bisector at the origin [2021 Fall AMC 12A — Problem 13]

The acute angle formed at the origin by the graphs of $y = x$ and $y = 3x$ has an angle bisector whose equation is $y = mx$.

Without using a calculator, determine the exact value of m .

Q4. Locus of the centroid — variable line through a fixed point [JEE Main — Straight Lines]

A variable line L passes through the fixed point (3, 5) and meets the positive x-axis at A and the positive y-axis at B.

Find the locus of the centroid G of triangle OAB, where O is the origin.
Identify the type of curve and sketch its key features.

Q5. Rotation of axes and preservation of a distance sum [JEE Main — Coordinate Geometry PYQ]

A line L has x-intercept a and y-intercept b on the standard coordinate axes.
The coordinate axes are now rotated through a fixed angle θ about the origin.
In the new coordinate system, the same line L has intercepts p and q .

Prove that: $1/a^2 + 1/b^2 = 1/p^2 + 1/q^2$

Q6. Midpoints of perpendicular segments between parallel lines [JEE Main 2021 — Straight Lines]

Find the locus of the midpoints of the perpendicular segments drawn from points on the line $x = 2y$ to the line $x = y$.

Interpret the result geometrically.

Q7. Concurrent line through centroid and intersection point [JEE Main 2020]

Let C be the centroid of the triangle with vertices (3, -1), (1, 3), and (2, 4).

Let P be the point of intersection of the lines $x + 3y - 1 = 0$ and $3x - y + 1 = 0$.

The line through C and P also passes through which of the following points?

- (a) $(-9, -6)$ (b) $(9, -6)$ (c) $(9, 6)$ (d) $(-9, 6)$

Q8. Fixed point of a two-parameter family of lines [JEE Advanced preparation — Straight Lines]

Consider the family of lines $x(a + 2b) + y(a + 3b) = a + b$, where a and b are arbitrary real parameters (not both zero).

- (a) Show that all lines in this family pass through a single fixed point P.
 (b) Find the coordinates of P.
 (c) Show that P lies on the angle bisector of the coordinate axes and determine which bisector.

Q9. Area of parallelogram enclosed by four lines [JEE Main — Coordinate Geometry PYQ]

Four lines $ax \pm by \pm c = 0$ (where a, b, c are positive constants) form a parallelogram.

- (a) Find the area of this parallelogram.
 (b) Show that the diagonals of this parallelogram are perpendicular (i.e. it is a rhombus).
 (c) Find the length of each diagonal in terms of a, b , and c .

Q10. Locus via distance-ratio condition to three lines [Adapted from HMMT / Competition Problem Sets]

In the coordinate plane, a point P moves such that the ratio of its perpendicular distance from the line $3x + 4y = 0$ to its perpendicular distance from the line $5x - 12y = 0$ equals 2.

- (a) Derive the equations of the locus of P.
 (b) Identify each branch of the locus and determine whether P can lie on both branches.
 (c) Find all points where the two branches intersect, if any.

SECTION 2 — PLANES (3D Analytical Geometry)

Problems in this section involve planes and lines in three-dimensional space. Topics include mirror images, coplanarity, foot-of-perpendicular loci, dihedral angles, families of planes, and distance relationships.

Problems from Top Institutions (Internet-Sourced)

Q1. Locus of feet of perpendiculars from a line to a plane [JEE Advanced — 3D Geometry PYQ]

Perpendiculars are drawn from every point on the line

$$(x + 2)/2 = (y + 1)/(-1) = z/3$$

to the plane $x + y + z = 3$.

Prove that the feet of all such perpendiculars are collinear, and find the equation of the line on which they lie.

Q2. Image of a point in a plane — then find the plane through image and a line [JEE Advanced — 3D Geometry PYQ]

Let P be the mirror image of the point $(3, 1, 7)$ with respect to the plane $x - y + z = 3$.

Find the equation of the plane that passes through P and contains the line $x/1 = y/2 = z/1$. Verify your answer by confirming the line lies entirely in the plane.

Q3. Plane through a point, perpendicular to two given planes [JEE Advanced — 3D Geometry PYQ]

Find the equation of the plane that passes through the point $(1, 1, 1)$ and is simultaneously perpendicular to both of the planes:

$$P_1: 2x + y - 2z = 5$$

$$P_2: 3x - 6y - 2z = 7$$

Give the answer in the form $ax + by + cz = d$ with integer coefficients.

Q4. Coplanarity of lines depending on a parameter — finding the intersection point [JEE Advanced — 3D Geometry PYQ]

For some $\alpha \in \mathbb{R}$, the following two lines are coplanar:

$$L_1: (x + 1)/2 = (y - 2)/(-1) = (z - 1)/1$$

$$L_2: (x + 2)/\alpha = (y + 1)/(5 - \alpha) = (z + 1)/1$$

- Determine α .
- Write the symmetric equations of L_2 for your value of α .
- Determine which of the following points lies on L_2 :
 $(2, -10, -2)$, $(10, -2, -2)$, $(10, 2, 2)$, $(-2, 10, 2)$

Q5. Distance from a point to a plane measured along a given direction [JEE Advanced — 3D Geometry PYQ]

Find the distance from the point $(1, -2, 3)$ to the plane $x - y + z = 5$,

measured parallel to the line $x/2 = y/3 = z/(-6)$.

Compare this to the true perpendicular distance and explain why they differ.

Q6. Mirror image via a plane — distance from an axis [JEE Advanced — 3D Geometry PYQ]

A point P lies in the first octant. Its mirror image Q in the plane $x + y = 3$ lies on the z-axis. The distance of P from the x-axis is 5.

Let R be the mirror image of P in the xy-plane.

- (a) Find the coordinates of P.
- (b) Find the length PR.
- (c) Verify that the triangle PQR is right-angled and identify the right angle vertex.

Q7. Mirror image of a point — recovering the plane from two symmetric points [JEE Advanced — 3D Geometry PYQ]

Given that $\alpha^2 + \beta^2 + \gamma^2 \neq 0$ and $\alpha + \gamma = 1$, the point $(3, 2, -1)$ is the mirror image of $(1, 0, -1)$ with respect to the plane $\alpha x + \beta y + \gamma z = \delta$.

Determine which of the following statements are TRUE:

- (a) $\alpha + \beta = 2$
- (b) $\delta - \gamma = 3$
- (c) $\delta + \beta = 4$
- (d) $\alpha + \beta + \gamma = \delta$

Justify each answer by recovering the plane equation explicitly.

Q8. Family of planes through an intersection line — distance conditions [JEE Advanced — 3D Geometry PYQ]

In $\mathbb{R}^3$, consider the planes $P_1: y = 0$ and $P_2: x + z = 1$.

Let P_3 be a plane (different from P_1 and P_2) passing through the line $P_1 \cap P_2$.

Given that the distance of the point $(0, 1, 0)$ from P_3 is 1, and the distance of a point (α, β, γ) from P_3 is 2,

determine which of the following relations must hold:

- (a) $2\alpha + \beta + 2\gamma + 2 = 0$
- (b) $2\alpha - \beta + 2\gamma + 4 = 0$
- (c) $2\alpha + \beta - 2\gamma - 10 = 0$
- (d) $2\alpha - \beta + 2\gamma - 8 = 0$

Q9. Coplanar lines — finding the containing plane(s) [JEE Advanced 2012]

If the two lines $(x - 1)/2 = (y + 1)/k = z/2$ and $(x + 1)/5 = (y + 1)/2 = z/k$ are coplanar, find all possible values of k and determine the plane(s) that contain both lines.

Q10. Two lines and a plane — mutual perpendicularity and intersection [JEE

Advanced — 3D Geometry (Two-Lines-and-Plane problems)]

Let ℓ_1 and ℓ_2 be the lines $\vec{r}_1 = \lambda(\hat{i} + \hat{j} + \hat{k})$ and $\vec{r}_2 = (\hat{j} - \hat{k}) + \mu(\hat{i} + \hat{k})$.

Let X be the set of all planes H that contain the line ℓ_1 .

- (a) For every $H \in X$, show that ℓ_2 and H are either parallel or $\ell_2 \subset H$.
- (b) Determine the distance from the point $(0, 1, 0)$ to the plane in X that contains ℓ_2 .
- (c) Determine the angle between ℓ_2 and the plane in X closest to the point $(0, 1, 0)$.

BRILLIANT-LEVEL — Original Problems

The following two problems are original compositions not found on the internet. They are crafted at the difficulty level of Brilliant.org's hardest analytical geometry challenges — requiring advanced synthesis of envelopes, projective reasoning, and three-dimensional metric geometry.

★ BRILLIANT Q1 Envelope of a One-Parameter Family — Tangents and Regions [Lines]

Consider the one-parameter family of lines in the Cartesian plane:

$$F(t) : (t^2 + 1)x - 2ty - (t^2 - 1) = 0, \quad t \in \mathbb{R}.$$

Answer all three parts below:

- (a) Show that no single point lies on every line of the family $F(t)$.
- (b) Find the envelope of the family $F(t)$. [The envelope is the curve tangent to every member of the family at some point.] Identify the conic section it defines and give its standard-form equation, center, and semi-axes (or focus-directrix pair).
- (c) Let $Q = (a, b)$ be an arbitrary point NOT on the envelope. Determine — as a function of a and b — exactly how many lines of the family $F(t)$ pass through Q . Then, for the case $Q = (3, 0)$, find those lines explicitly and verify that the chord of contact of the two tangents from Q to the envelope has the equation predicted by the standard polar-of-a-point formula.

★ BRILLIANT Q2 Orthogonal Triplet of Planes and an Inscribed Sphere [Planes]

Let Π_1, Π_2, Π_3 be three mutually perpendicular planes in $\mathbb{R}^3$, all passing through the origin $O = (0, 0, 0)$. Denote by $\hat{n}_1, \hat{n}_2, \hat{n}_3$ their respective unit normal vectors, so that $\{\hat{n}_1, \hat{n}_2, \hat{n}_3\}$ forms an orthonormal basis of $\mathbb{R}^3$.

A sphere Σ of radius $r > 0$ is tangent to all three planes Π_1, Π_2, Π_3 from the same half-space (i.e. its centre C lies on the positive-normal side of every plane).

Answer all three parts below:

- (a) Let $P = (x_0, y_0, z_0)$ be any point in $\mathbb{R}^3$, and let P_1, P_2, P_3 be its orthogonal projections onto Π_1, Π_2, Π_3 respectively.
 Prove that: $|OP_1|^2 + |OP_2|^2 + |OP_3|^2 = 2(x_0^2 + y_0^2 + z_0^2)$
 regardless of the orientation of the three planes — i.e. regardless of the choice of orthonormal basis $\{\hat{n}_1, \hat{n}_2, \hat{n}_3\}$. (Hint: use the decomposition of OP in the basis.)
- (b) Prove that the centre C of the sphere Σ satisfies $|OC| = r\sqrt{3}$, and hence lies on a sphere of radius $r\sqrt{3}$ centred at O .
 [Hint: the distance from C to each plane equals r , and use part (a) or direct computation via the orthonormal basis.]
- (c) As the three planes Π_1, Π_2, Π_3 rotate rigidly about O (maintaining mutual perpendicularity and the common point O), and the sphere Σ adjusts to remain tangent to all three from the same side, prove that the locus of the centre C is exactly the sphere $x^2 + y^2 + z^2 = 3r^2$. Find the eight possible positions of C (corresponding to the eight octants) when the coordinate planes are chosen as $\Pi_1 = \{y=0\}$, $\Pi_2 = \{x=0\}$, $\Pi_3 = \{z=0\}$, and verify they all satisfy $|OC| = r\sqrt{3}$.

Sources

- [1] JEE Advanced 3D Geometry PYQ (BYJU'S compilation)
<https://byjus.com/jee/3-dimension-jee-advanced-previous-year-questions-with-solutions/>
- [2] JEE Main Coordinate Geometry PYQ (BYJU'S compilation)
<https://byjus.com/jee/jee-main-maths-coordinate-geometry-previous-year-questions-with-solutions/>
- [3] JEE Main Straight Lines PYQ (BYJU'S compilation)
<https://byjus.com/jee/jee-straight-line-previous-year-questions-with-solutions/>
- [4] JEE Advanced Straight Lines PYQ (BYJU'S compilation)
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[5] 2021 Fall AMC 12A Problems — AoPS Wiki

https://artofproblemsolving.com/wiki/index.php/2021_Fall_AMC_12A_Problems/Problem_13

[6] eSaral — JEE Advanced 3D Geometry Previous Year Questions

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[7] ExamSide — JEE Advanced Mathematics: 3D Geometry

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[9] Vedantu — Three-Dimensional Geometry JEE Main Important Questions

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[10] Matherama — Angle Bisectors (Analytical Geometry)

<https://matherama.com/Tutorials/Chapter%2008%20-%20Straight%20Lines/07%20-%20Angle%20Bisectors/>

The two Brilliant-level problems at the end of this document are original compositions and have no internet source.